Degala Chenchupradeep

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PROFESSIONAL SUMMARY

Ph.D. Researcher in Quantum Machine Learning with expertise in AI, Deep Learning, and Quantum Computing. Proficient in Python, Qiskit, and PennyLane. Develops quantum-inspired and gate-based models for many-body physics, error mitigation, and QAOA optimization, with applied work in quantum communication (QKD), underwater acoustic networks, and quantum-like fuzzy decision making. Ph.D. Scholar at SRM University AP.

CORE SKILLS

- **Quantum Computing:** Qiskit, PennyLane, Cirq, Tensor Networks (MPS), ZNE (Error Mitigation), VQA, QAOA, VQE.
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Reinforcement Learning, NLP (BERT), GCN.
- **Tools & Languages:** Python, C++, SQL, Django, Flask, Pandas, NetworkX, Big Data (Spark, Hadoop), Cloud Computing.

RECENT RESEARCH PROJECTS (OPEN SOURCE)

- **Chip-aware QAOA placement on modular quantum devices** *GitHub: modular-chip-aware-qaoa-placement*
  – Kernighan-Lin chip placement raised noisy QAOA MaxCut approximation ratio by 0.04-0.06 over random placement at 20x coupler error (n=8-12, simulation); zero gain at equal error.
- **Break-even noise for constrained XY-mixer vs penalty QAOA** *GitHub: xy-mixer-vs-penalty-qaoa-noise-crossover*
  – Pauli-trajectory simulation (n=8, 5 graphs): depth-2 XY-Dicke QAOA drops from 0.758 to 0.390 feasible-weighted ratio at p2=5e-3, below penalty QAOA noiseless 0.349 by ~1e-2.
- *+1 further reproducible repositories at github.com/pradeepdegalaroyal.*

PH.D. & CORE PROJECTS (PORTFOLIO)

- **Hardware-Efficient-QPU-Ansatz | VQE for H_2** *GitHub: Hardware-Efficient-QPU-Ansatz*
  – Implemented VQE for the hydrogen molecule with a hardware-efficient ansatz tailored to Rigetti-style QPU topology.
  – Used native gate sets (Ry, Rz) and nearest-neighbor CNOT constraints; converged to -1.114 Hartree.
- **QEM-ZeroNoise-Extrapolation | Quantum Error Mitigation** *GitHub: QEM-ZeroNoise-Extrapolation*
  – Developed ZNE protocols using circuit folding and Richardson extrapolation on a custom density-matrix simulator.
  – Reduced expectation-value error from 31.5% to 5.1% under simulated depolarizing noise models.
- **QML-TensorNet-ManyBody | MPS Simulation** *GitHub: QML-TensorNet-ManyBody*
  – Built a Matrix Product State (MPS) solver from scratch to simulate the 1D Transverse-Field Ising Model.
  – Analyzed the quantum phase transition at $h = 1.0$, matching exact diagonalization for many-body systems.
- **VQA-Quantum-Dynamics | Trotterized Simulation** *GitHub: VQA-Quantum-Dynamics*
  – Translated Heisenberg XY Hamiltonian dynamics into Trotterized gate circuits using PennyLane.
  – Simulated magnon propagation in a 4-qubit spin chain starting from a Néel state.
- **QML-Advantage-Benchmarking | QAOA vs Classical** *GitHub: QML-Advantage-Benchmarking*
  – Benchmarked QAOA against brute-force MaxCut solvers to define the "crossover regime".
  – Quantified approximation ratios and runtime scaling across graph sizes ($N = 4$ to $N = 10$).

PROFESSIONAL EXPERIENCE

Assistant Professor, Computer Science | Raghu Engineering College *Aug 2024 – Nov 2024*

- Taught AI/ML courses; mentored undergraduate projects in quantum and data science; developed new curriculum.

Backend Web Development Intern | PMN Company/Chegg *Jun 2023 – Aug 2023*

- Built Django/Flask services for education platforms, integrated REST APIs, and optimized database queries.

EDUCATION

- **Ph.D. in Computer Science (Pursuing)** | SRM University AP, India *Ongoing*
  – *Research Focus:* Quantum Machine Learning and QAOA Optimization.
- **M.Tech in Data Science (Gold Medalist)** | JNTU-Gurajada *2022 – 2024*
- **B.Tech in Mechanical Engineering** | Siddharth Institute of Engineering *2019 – 2022*

CERTIFICATIONS & HONORS

- **Gold Medalist** in M.Tech Data Science for academic excellence.
- **Quantum Algorithms & Cryptography** | NPTEL (IIT Madras).
- **Deep Learning** | NPTEL.
- **Python for Data Science** | NPTEL.